USN

[22AN552]

# FIFTH SEMESTER B.Tech EXAMINATION, DECEMBER - 2024

## AIR TRAFFIC MANAGEMENT

**Time: 3 Hours** **Maximum Marks: 100**

**Instructions:** i. Missing data may be suitably assumed.

**ANSWER ALL QUESTIONS**

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| | | **PART-A** | **5 x2=10** | | |
| 1. | a. | Classify Approach Control Service. | L2 | CO1 | 02 |
| | b. | Describe the role of air traffic control instructions in maintaining separation. | L2 | CO2 | 02 |
| | c. | Interpret the Annexure of the Chicago Convention which deals with Air Traffic Services. | L3 | CO3 | 02 |
| | d. | Interpret Restricted flying zones. | L3 | CO4 | 02 |
| | e. | Explain the concept of Cabotage. | L2 | CO5 | 02 |
| | | **PART-B** | **6 X 5=30** | | |
| 2. | a. | Classify vertical and horizontal speed control instructions. | L2 | CO1 | 05 |
| | | **OR** | | | |
| | b. | Recognize the various types of air traffic control clearances. | L2 | CO1 | 05 |
| 3. | a. | Explain the concept of horizontal separation and its minimum requirements. | L2 | CO2 | 05 |
| | | **OR** | | | |
| | b. | Recognize the fundamental principles of separation in air traffic management. | L2 | CO2 | 05 |
| 4. | a. | Interpret VMC and IMC. | L3 | CO3 | 05 |
| | | **OR** | | | |
| | b. | Demonstrate the differences between VFR and IFR. | L3 | CO3 | 05 |
| 5. | a. | Interpret the following METAR- VABB 050800Z 04005KT 3000 FU NSC 35/14 Q1010 NOSIG | L3 | CO4 | 05 |
| | | **OR** | | | |
| | b. | Interpret the following METAR- VOBL 050800Z 06008KT 8000 SCT012 SCT020 27/16 Q1015 NOSIG | L3 | CO4 | 05 |
| 6. | a. | Explain TCAS and its importance. | L2 | CO5 | 05 |
| | | **OR** | | | |
| | b. | Recognize runway visual range to assess the suitability of a runway for landing in low-visibility conditions. | L2 | CO5 | 05 |

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| **7.** | **a.** | Interpret precision approach of an aircraft. | L3 | CO6 | **05** |
| | | **OR** | | | |
| | **b.** | Demonstrate non-precision approach of an aircraft. | L3 | CO6 | **05** |
| | | | | | |
| | | **PART-C** | | **6 X 10=60** | |
| **8.** | **a.** | Interpret the differences between primary radar and secondary radar, and its accuracy. | L3 | CO3 | **10** |
| | | **OR** | | | |
| | **b.** | Interpret the role of receiver controls to optimize radar performance in different weather conditions. | L3 | CO3 | **10** |
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| **9.** | **a.** | Demonstrate the different Air Traffic Regions in India and their importance. | L3 | CO4 | **10** |
| | | **OR** | | | |
| | **b.** | Interpret "Urgency" and "Emergency" calls made by aircraft with suitable examples. | L3 | CO4 | **10** |
| | | | | | |
| **10.** | **a.** | Explain airport emergency procedures to respond to an aircraft accident or incident. | L2 | CO5 | **10** |
| | | **OR** | | | |
| | **b.** | Interpret the approach control procedures to optimize the efficiency of aircraft arrivals and departures. | L2 | CO5 | **10** |
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| **11.** | **a.** | Identify the challenges associated with maintaining separation in remote or uncontrolled airspace. | L2 | CO2 | **10** |
| | | **OR** | | | |
| | **b.** | Explain the factors that influence the determination of separation minima. | L2 | CO2 | **10** |
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| **12.** | **a.** | Classify the different types of Freedom given to commercial aircraft around the world. | L2 | CO1 | **10** |
| | | **OR** | | | |
| | **b.** | Describe the functions and objectives of Air Traffic Services. | L2 | CO1 | **10** |
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| **13.** | **a.** | Demonstrate the various air traffic control procedures to improve the safety and efficiency of air traffic operations. | L3 | CO6 | **10** |
| | | **OR** | | | |
| | **b.** | Interpret the responsible division of ATC for different stages of the flight. | L3 | CO6 | **10** |